O'rtacha tezlik

Avtobus shahar ichida sekin, katta yo'lda tez yuradi. Butun yo'lda uning tezligi qancha bo'lgan? Bunday savolga o'rtacha tezlik javob beradi.

1-QISM

Mavzu

ESLAB OLAYLIK

- Tezlik = masofa : vaqt; masofa = tezlik · vaqt; vaqt = masofa : tezlik.
- 30 min = $\frac{1}{2}$ soat, 15 min = $\frac{1}{4}$ soat, 20 min = $\frac{1}{3}$ soat, 45 min = $\frac{3}{4}$ soat.
- Aralash songa bo'lish: $90 : 1\frac{1}{2} = 90 : \frac{3}{2} = 90 \cdot \frac{2}{3} = 60.$

QOIDA · O'RTACHA TEZLIK

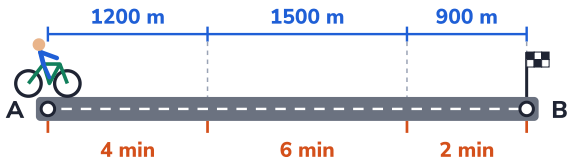
O'rtacha tezlik = jami masofa : jami vaqt



Avtobus goh tez, goh sekin yuradi. Agar u butun yo'lni o'rtacha tezlikda bir tekis yursa, manzilga xuddi shu vaqtda yetib borardi.

1-MISOL · UCH QISM

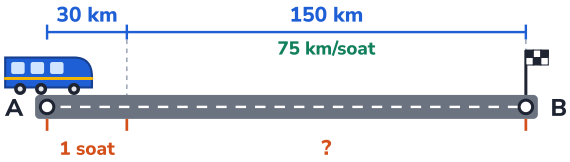
Velosipedchi bog' yo'lida to'xtamasdan uch qism yo'l yurdi: 1200 m ni 4 minutda, 1500 m ni 6 minutda, 900 m ni 2 minutda. Uning o'rtacha tezligi qancha?



1. Jami masofa: $1200 + 1500 + 900 = 3600$ m.
2. Jami vaqt: $4 + 6 + 2 = 12$ min.
3. O'rtacha tezlik: $3600 : 12 = 300$ m/min.

2-MISOL · BIR QISMDA TEZLIK BERILGAN

Elektropoyezd shahar ichida 30 km ni 1 soatda yurdi. Shahardan chiqqach, 150 km ni 75 km/soat tezlikda yurdi. Uning butun yo'ldagi o'rtacha tezligi qancha?



1. Ikkinchi qism vaqti: $150 : 75 = 2$ soat.
2. Jami masofa: $30 + 150 = 180$ km. Jami vaqt: $1 + 2 = 3$ soat.
3. O'rtacha tezlik: $180 : 3 = 60$ km/soat.

3-MISOL · JAVOB BUTUN EMAS

Sportchi stadionda ikki aylana yugurdi. Bir aylana 400 m. Birinchi aylanani 60 sekundda, ikkinchisini 65 sekundda yugurdi.

a) U jami qancha vaqt yugurdi?

b) Uning o'rtacha tezligini toping.

a. Jami vaqt: $60 + 65 = 125$ s, ya'ni 2 min 5 s.

b. Jami masofa: $400 + 400 = 800$ m. O'rtacha tezlik: $800 : 125 = 6$ (qoldiq 50), ya'ni $6\dfrac{50}{125} = 6\dfrac{2}{5}$ m/s yoki $6{,}4$ m/s.

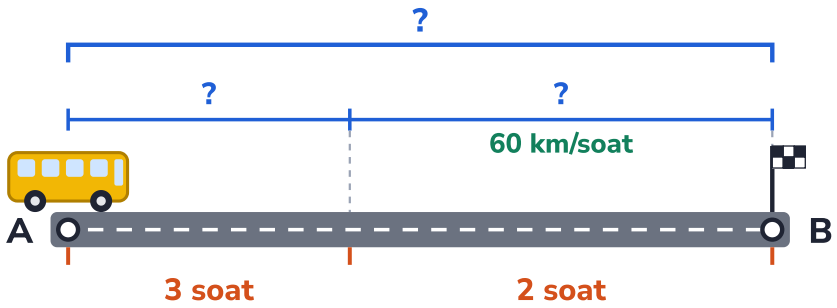
4-MISOL · O'RTACHA TEZLIK MA'LUM

Samolyot A shahardan C shaharga B shahar ustidan qo'nmasdan uchdi. A dan B gacha 1500 km ni 2 soatda uchdi, B dan C gacha yana 1650 km uchdi. Butun yo'ldagi o'rtacha tezligi 700 km/soat. B dan C gacha uchishga qancha vaqt ketgan?

1. Jami masofa: $1500 + 1650 = 3150$ km.
2. Jami vaqt = jami masofa : o'rtacha tezlik: $3150 : 700 = 4\dfrac{1}{2}$ soat.
3. B dan C gacha: $4\dfrac{1}{2} - 2 = 2\dfrac{1}{2}$ soat, ya'ni 2 soat 30 min.

5-MISOL · YO'LNING QISMI BERILGAN

Sayyohlar avtobusi yo'lning $\frac{2}{5}$ qismini tog'da 3 soatda bosib o'tdi. Qolgan qismini tekislikda 60 km/soat tezlikda 2 soatda yurdi. Avtobusning butun yo'ldagi o'rtacha tezligi qancha?



1. Qolgan qism — yo'lning $1 - \frac{2}{5} = \frac{3}{5}$ qismi. Uning uzunligi: $60 \cdot 2 = 120$ km.
2. $\frac{3}{5}$ qism 120 km bo'lsa, $\frac{1}{5}$ qism $120 : 3 = 40$ km. Butun yo'l: $40 \cdot 5 = 200$ km.
3. Jami vaqt: $3 + 2 = 5$ soat. O'rtacha tezlik: $200 : 5 = 40$ km/soat.

O'rtacha tezlik formulasidan yana ikkitasi chiqadi: jami masofa = o'rtacha tezlik · jami vaqt; jami vaqt = jami masofa : o'rtacha tezlik.

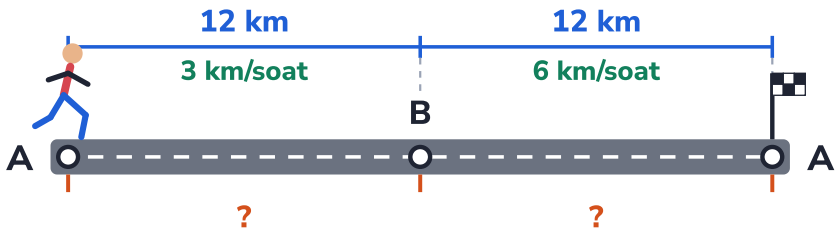


2-QISM

Misollar ishlash

1-MISOL · BORIB KELISH

Sayyohlar A lagerdan B ko'lgacha 12 km yo'lni tepalikka 3 km/soat tezlikda ko'tarilib bosib o'tdi. Xuddi shu yo'ldan 6 km/soat tezlikda pastga tushib qaytdi. Butun yo'ldagi o'rtacha tezlik qancha?

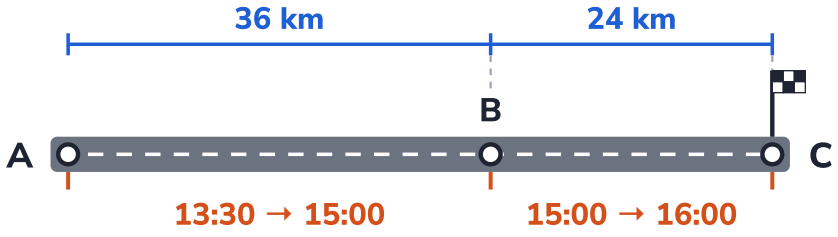


1. Ko'tarilish vaqti: $12 : 3 = 4$ soat. Tushish vaqti: $12 : 6 = 2$ soat.
2. Jami masofa: $12 + 12 = 24$ km (yo'l ikki marta bosildi). Jami vaqt: $4 + 2 = 6$ soat.
3. O'rtacha tezlik: $24 : 6 = 4$ km/soat.

Tezliklarning o'rtachasi $(3 + 6) : 2 = 4{,}5$ km/soat — bu javob noto'g'ri. Sayyohlar 4 soat sekin, faqat 2 soat tez yurdi. Sekin yurilgan vaqt ko'p bo'lgani uchun o'rtacha tezlik kichikroq chiqdi.

2-MISOL · VAQT SOATLAR BILAN BERILGAN

Paroxod soat 13:30 da A pristandan jo'nadi. 15:00 da B pristanga yetdi va to'xtamasdan suzib, 16:00 da C pristanga yetdi. A dan B gacha 36 km, B dan C gacha 24 km. Paroxodning o'rtacha tezligi qancha?



1. Jami vaqt: 13:30 dan 16:00 gacha — 2 soat 30 min = $2\frac{1}{2}$ soat = 2,5 soat.

2. Jami masofa: $36 + 24 = 60$ km.
3. O'rtacha tezlik: $60 : 2{,}5 = 600 : 25 = 24$ km/soat.

Agar savol boshqacha bo'lsa: «o'rtacha tezligi 24 km/soat, 13:30 dan 16:00 gacha suzdi — qancha yo'l bosdi?» — unda jami masofa = o'rtacha tezlik · jami vaqt: $24 \cdot 2{,}5 = 60$ km.

3-MISOL · JADVAL BILAN

Velosipedchi va mopedchi P qishloqdan R qishloqqa Q qishloq orqali bordi. P dan Q gacha 20 km, Q dan R gacha 25 km. Jadvalni to'ldiring.

	P dan Q gacha	Q dan R gacha	Jami masofa	Jami vaqt	O'rtacha tezlik
Velosipedchi	$1\frac{1}{2}$ soat	$1\frac{1}{2}$ soat	45 km	3 soat	15 km/soat
Mopedchi	$\frac{1}{2}$ soat	1 soat	45 km	$1\frac{1}{2}$ soat	30 km/soat

1. Jami masofa ikkalasida ham bir xil: $20 + 25 = 45$ km.
2. Velosipedchi: $1\frac{1}{2} + 1\frac{1}{2} = 3$ soat, $45 : 3 = 15$ km/soat.
3. Mopedchi: $\frac{1}{2} + 1 = 1\frac{1}{2}$ soat, $45 : 1\frac{1}{2} = 45 \cdot \frac{2}{3} = 30$ km/soat.



Borib kelishda yo'l ikki marta bosiladi: ko'lgacha 12 km bo'lsa, borib kelish — 24 km.

DIQQAT!

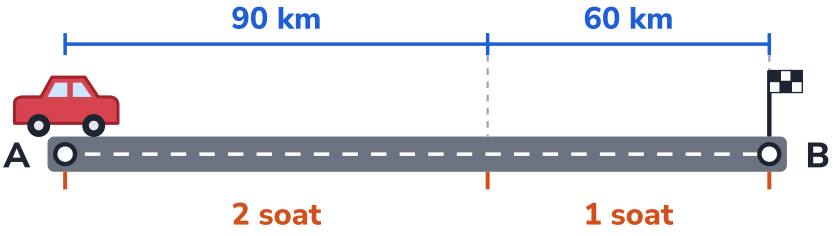
1. Tezliklarning o'rtachasini olmang. 12 km ni 3 km/soat, qaytishda 6 km/soat: ✗ $(3+6):2=4{,}5$ km/soat ✓ $24:6=4$ **km/soat**
2. Tezligi berilgan qismning vaqtini avval toping, keyin hamma vaqtlarni qo'shing. 30 km — 1 soatda, 150 km — 75 km/soat tezlikda: ✗ faqat 1 soatni olish: $180:1=180$ ✓ $180:(1+2)=60$ **km/soat**
3. Minutlarni soatga to'g'ri aylantiring: ✗ 2 soat 30 min = 2,3 soat ✓ **2 soat 30 min = 2,5 soat**
4. O'rtacha tezlik eng kichik va eng katta tezlik orasida bo'ladi. 3 km/soat va 6 km/soat yurilgan bo'lsa, o'rtacha tezlik 3 dan katta, 6 dan kichik.

3-QISM

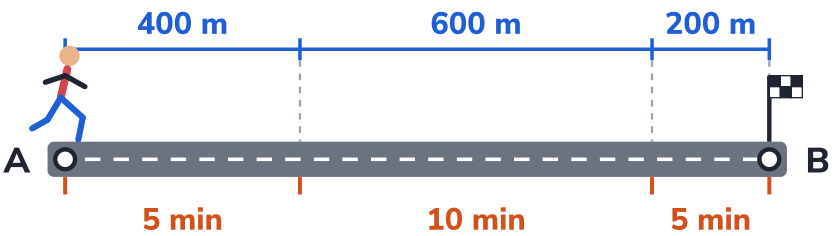
Mustahkamlash

1. Rasmga qarab o'rtacha tezlikni toping.

   a) Mashina:



   b) Piyoda (har qismda masofa va vaqt):



2. Uch kishi A qishloqdan C qishloqqa B qishloq orqali bordi. A dan B gacha 12 km, B dan C gacha 18 km. Jadvalni daftaringizga chizib to'ldiring.

	A dan B gacha	B dan C gacha	Jami masofa	Jami vaqt	O'rtacha tezlik
Velosipedchi	1 soat	1 soat	?	?	?
Samokatchi	$\frac{1}{2}$ soat	1 soat	?	?	?
Otliq	$\frac{1}{2}$ soat	$\frac{3}{4}$ soat	?	?	?

3. Elektrsamokatchi $\frac{1}{4}$ soatda 4 km, keyin $\frac{3}{4}$ soatda 8 km yurdi. Uning o'rtacha tezligini toping.

4. Qayiq ko'ldagi orolga borib keldi. Qirg'oqdan orolgacha 9 km. Qayiqning borib kelishdagi o'rtacha tezligi 12 km/soat. Qayiq jami qancha vaqt suzdi?

5. Chang'ichi tepalikka olib chiqadigan 2 km yo'lni $\frac{1}{2}$ soatda bosib o'tdi. Xuddi shu yo'ldan $\frac{1}{6}$ soatda (10 minutda) pastga tushdi. Chang'ichining butun yo'ldagi o'rtacha tezligi qancha?

6. Sayyohlar avtobusining safari $4\frac{1}{2}$ soat davom etdi. Avtobus avval tekis yo'lda 65 km/soat tezlikda 2 soat yurdi, so'ng 95 km lik tog' yo'lini bosib o'tdi.
   a) Avtobus jami qancha yo'l bosdi?
   b) Uning o'rtacha tezligi qancha?

7. Teploxod daryoda 108 km yo'lni 36 km/soat tezlikda suzib o'tdi. Bundan oldin u 3 soat suzgan edi. Teploxodning butun yo'li 180 km.
   a) Teploxod jami qancha vaqt suzdi?
   b) Uning o'rtacha tezligi qancha?

8. Traktor fermadan dalaga 4 soatda yetib bordi. Oxirgi 1 soatda u 12 km/soat tezlikda yurdi va shu vaqtda yo'lning $\frac{1}{3}$ qismini bosdi.
   a) Fermadan dalagacha necha km?
   b) Traktorning o'rtacha tezligi qancha?

9. Ikki bekat orasi 280 km, yo'lning o'rtasida yana bir bekat bor. Poyezd o'rtadagi bekatgacha $1\frac{1}{2}$ soat yurdi va u yerdan to'xtamasdan o'tib, qolgan yo'lni 56 km/soat tezlikda bosdi.
   a) Poyezd butun yo'lga qancha vaqt sarfladi?
   b) Poyezdning o'rtacha tezligi qancha?

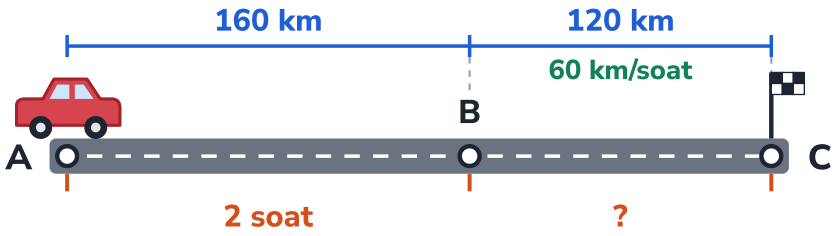
10. Kater 90 km lik yo'lning 60 km ini 40 km/soat tezlikda suzdi. Qolgan yo'lni 1 soatda suzdi. Katerning butun yo'ldagi o'rtacha tezligini toping.

11. Sayyohlar poyezdi 09:40 da jo'nab, 11:10 da birinchi bekatga yetdi. To'xtamasdan yo'lini davom ettirib, 12:10 da manzilga yetdi. Poyezdning o'rtacha tezligi 64 km/soat.
    a) Poyezd jami qancha yo'l bosdi?
    b) Birinchi bekatgacha 90 km bo'lsa, poyezd ikkinchi qismda qanday tezlikda yurgan?

4-QISM

Mustaqil ish

1. Rasmdagi yo'l uchun mashinaning o'rtacha tezligini toping.



2. Suzuvchi hovuzda 150 m suzdi: birinchi 50 m ni 40 sekundda, keyingi 50 m ni 50 sekundda, oxirgi 50 m ni 60 sekundda. Uning o'rtacha tezligi necha m/s?

3. Velosipedchi uyidan bozorga 15 km/soat tezlikda 1 soatda yetib bordi. Qaytishda qarshi shamol esgani uchun 10 km/soat tezlikda yurdi. Velosipedchining borib kelishdagi o'rtacha tezligi qancha?

4. Avtobus avval 50 km ni 1 soatda yurdi, keyin yana 160 km yurdi. Butun yo'ldagi o'rtacha tezligi 70 km/soat. Avtobus ikkinchi qismga qancha vaqt sarflagan?

5. Kema yo'lning $\frac{1}{4}$ qismini 2 soatda suzdi. Qolgan qismini 30 km/soat tezlikda 3 soatda suzdi. Kemaning o'rtacha tezligini toping.

6. Dron 10:05 da uchib, 10:25 gacha 6 km uchdi. So'ng 10:35 gacha yana 4 km uchdi. Dronning o'rtacha tezligi necha km/soat?

7. **Xatoni toping.** Mashina 90 km yo'lni 60 km/soat tezlikda bosib o'tib, shu yo'ldan 30 km/soat tezlikda qaytdi. Nodira o'rtacha tezlikni shunday topdi: $(60 + 30) : 2 = 45$ km/soat. Xato nimada? To'g'ri javob qancha?

8. **Qiyinroq.** Velosipedchi 2 soat 15 km/soat tezlikda yurdi, keyin yana 1 soat yurdi. Uning 3 soatdagi o'rtacha tezligi 14 km/soat bo'ldi. Oxirgi soatda u qanday tezlikda yurgan?

5-QISM

Javoblar

Mustahkamlash

1-masala

a) $90 + 60 = 150$ km, $2 + 1 = 3$ soat, $150 : 3 = 50$ km/soat. b) $400 + 600 + 200 = 1200$ m, $5 + 10 + 5 = 20$ min, $1200 : 20 = 60$ m/min.

2-masala

Jami masofa hammasida $12 + 18 = 30$ km. Velosipedchi: 2 soat, $30 : 2 = 15$ km/soat. Samokatchi: $1\frac{1}{2}$ soat, $30 : 1\frac{1}{2} = 20$ km/soat. Otliq: $\frac{1}{2} + \frac{3}{4} = 1\frac{1}{4}$ soat, $30 : 1\frac{1}{4} = 30 \cdot \frac{4}{5} = 24$ km/soat.

3-masala

$4 + 8 = 12$ km, $\frac{1}{4} + \frac{3}{4} = 1$ soat. $12 : 1 = 12$ km/soat.

4-masala

Borib kelish: $9 + 9 = 18$ km. Vaqt: $18 : 12 = 1\frac{1}{2}$ soat, ya'ni 1 soat 30 min.

5-masala

$2 + 2 = 4$ km, $\frac{1}{2} + \frac{1}{6} = \frac{2}{3}$ soat. $4 : \frac{2}{3} = 4 \cdot \frac{3}{2} = 6$ km/soat.

6-masala

a) Tekis yo'lda: $65 \cdot 2 = 130$ km. Jami: $130 + 95 = 225$ km. b) $225 : 4\frac{1}{2} = 225 \cdot \frac{2}{9} = 50$ km/soat.

7-masala

a) $108 : 36 = 3$ soat; jami $3 + 3 = 6$ soat. b) $180 : 6 = 30$ km/soat.

8-masala

a) Oxirgi 1 soatda: $12 \cdot 1 = 12$ km — bu yo'lning $\frac{1}{3}$ qismi. Butun yo'l: $12 \cdot 3 = 36$ km. b) Jami vaqt 4 soat, $36 : 4 = 9$ km/soat.

9-masala

a) Yo'lning yarmi: $280 : 2 = 140$ km. O'rtadagi bekatdan keyin: $140 : 56 = 2\frac{1}{2}$ soat. Jami: $1\frac{1}{2} + 2\frac{1}{2} = 4$ soat. b) $280 : 4 = 70$ km/soat.

10-masala

Birinchi qism: $60 : 40 = 1\frac{1}{2}$ soat. Jami vaqt: $1\frac{1}{2} + 1 = 2\frac{1}{2}$ soat. $90 : 2\frac{1}{2} = 90 \cdot \frac{2}{5} = 36$ km/soat.

11-masala

a) 09:40 dan 12:10 gacha — 2 soat 30 min = $2\frac{1}{2}$ soat. $64 \cdot 2\frac{1}{2} = 160$ km. b) Ikkinchi qism: $160 - 90 = 70$ km, 11:10 dan 12:10 gacha — 1 soat. Tezligi 70 km/soat.

Mustaqil ish

1-masala

B dan C gacha: $120 : 60 = 2$ soat. Jami: $160 + 120 = 280$ km, $2 + 2 = 4$ soat. $280 : 4 = 70$ km/soat.

2-masala

$40 + 50 + 60 = 150$ s. $150 : 150 = 1$ m/s.

3-masala

Bozorgacha: $15 \cdot 1 = 15$ km. Qaytish: $15 : 10 = 1\frac{1}{2}$ soat. Jami: 30 km, $2\frac{1}{2}$ soat. $30 : 2\frac{1}{2} = 12$ km/soat (12,5 emas).

4-masala

Jami masofa: $50 + 160 = 210$ km. Jami vaqt: $210 : 70 = 3$ soat. Ikkinchi qism: $3 - 1 = 2$ soat.

5-masala

Qolgan $\frac{3}{4}$ qism: $30 \cdot 3 = 90$ km. $\frac{1}{4}$ qism: $90 : 3 = 30$ km, butun yo'l $30 \cdot 4 = 120$ km. Vaqt: $2 + 3 = 5$ soat. $120 : 5 = 24$ km/soat.

6-masala

10:05 dan 10:35 gacha — 30 min = $\frac{1}{2}$ soat. Masofa: $6 + 4 = 10$ km. $10 : \frac{1}{2} = 20$ km/soat.

7-masala

Nodira tezliklarning o'rtachasini olgan. Borish: $90 : 60 = 1\frac{1}{2}$ soat, qaytish: $90 : 30 = 3$ soat. Jami: 180 km, $4\frac{1}{2}$ soat. $180 : 4\frac{1}{2} = 40$ km/soat.

8-masala

3 soatda jami: $14 \cdot 3 = 42$ km. Dastlabki 2 soatda: $15 \cdot 2 = 30$ km. Oxirgi soatda: $42 - 30 = 12$ km, ya'ni 12 km/soat.